SoK: On the Design of Secure Ranking-based E-voting Systems

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Abstract: Vote privacy and verifiability are two fundamental properties which must be maintained for elections. Studies in e-voting technology have demonstrated how elections may be secured through cryptographic techniques including mix-nets, homomorphic encryption, and Zero Knowledge Proofs (ZKPs). The implications and trade-offs that must be considered when designing an e-voting system using multiple cryptographic techniques for simple majority or plurality voting are well understood in the literature. However, it is less clear how these techniques affect the design process for complex e-voting systems such as those based on ranked voting. Examples include Borda count, Condorcet voting, Instant Runoff Voting (IRV), and Single Transferable Vote (STV). In this paper, we systematically analyse the cryptographic design of ranking-based e-voting systems in the literature and explore how the cryptographic techniques utilised by such systems affect the strength of their security guarantees including vote privacy and verifiability. In addition, we explore how their cryptographic designs impact social choice properties including support for various counting functions as well as flexibility in vote casting. Our findings demonstrate that ranked e-voting systems may be largely classified according to the degree of information leakage present in each solution, and that systems belonging to each of these categories share similar properties and usage of cryptographic techniques. We then discuss an overview of our findings and identify a number of open problems following our analysis. Overall, this paper serves as a comprehensive study on the state-of-the-art for secure ranked e-voting systems in the literature.

Keywords: E2E Verifiability, Vote Privacy, E-voting, Ranked Voting

1 Introduction

Many voting methods have been studied and applied to elections throughout history in order to provide results which are fair and representative with respect to the collective opinions of the voters. Arguably the simplest voting procedure for elections consisting of two or more choices is *plurality voting*, where a choice must receive more votes than any of the other choices to win the election. Plurality voting, whilst straightforward, is known to be one of the least representative voting procedures in social choice theory. It is particularly susceptible to the problem of *vote splitting*. This refers to the phenomenon where multiple choices have similar appeal amongst the group of voters, causing the group to divide their votes amongst the similar choices. This division of votes may then result in an overall less

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socially-desirable choice winning the election [Se20]. Vote splitting may be considered an example of *strategic voting*, the more general phenomenon where a voter chooses to cast their vote for a choice that is different to their favourite choice. The voter may attempt to prevent their strongly disliked choice from winning the plurality, or may decide to cast their vote for a choice that is allied to their favourite choice, in order to increase the likelihood that the election result is a coalition between both of them [BM96].

The shortcomings of plurality voting have long been acknowledged in the literature for social choice theory. Many voting methods have been considered in attempts to avoid vote splitting and lessen the impact of strategic voting on the election result. *Ranked voting* is a popular subset of these voting methods that requires the voters to provide more information regarding their opinions between the choices in an election in comparison to what is asked of them for plurality voting. The voters must list all or some of the choices in an order of their liking, and cast the resulting list as their vote. The lists of all voters are then taken into account when determining the election result. Example ranked methods include Borda count, Condorcet voting, Single Transferable Vote (STV), and Instant Runoff Voting (IRV). All of these have seen adoption for real-world elections [FG14; Ma03; Sc11; Sc18; TUK22].

Whilst many voting methods have been developed for elections throughout history, it was the advent of the digital era which brought about changes to the ways in which votes can be cast and tallied in an election. Traditional voting is often paper-based, requiring each voter to mark or write down their vote onto a paper ballot, which must then be deposited into a sealed ballot box. Later, once all votes have been cast, the seal on the ballot box is broken and the votes are counted to form a tally that determines the winner of the election. The manual balloting and counting procedures can be error-prone and tedious; this is especially the case for ranked voting, which is more complex than simple plurality voting. In order to improve upon the efficiency and fallibility of traditional voting, election organisers looked towards digital technologies to assist voters in casting their votes and speed up the counting procedures whilst avoiding the human factor of error. However, the use of digital technologies introduced new problems: the possibility that digital data can be easily read or altered to produce a different election result.

To overcome the threats to digital elections, researchers looked to provide systems which not only protect the privacy of each individual vote, but to also provide assurances on the correctness of the election result in a manner that enables anyone to verify the outcome of an election. Many of these e-voting solutions target ranked methods and look to at least maintain the privacy of the votes and verifiability of the result. However, the challenges which must be faced when designing a secure e-voting solution for ranked voting are considerably different to those which must be faced for plurality voting. In particular, these challenges differ in two aspects, which we outline as follows.

- *Privacy definitions*: The requirements to achieve vote privacy are different for ranked methods. It is not sufficient to only obscure the votes that were cast in an election, as

even revealing a portion of the final tally can be enough for a voter's privacy to be broken in a ranked election;

- *Architectural differences*: Existing e-voting solutions for ranked voting can differ in the voting methods that they support. Some solutions are specialised to a single voting method, whilst others aim to support a variety of different ones. The solutions that support many ranked methods often differ in the techniques used in comparison to those that only support a single ranked method. Where these techniques are shared between solutions, the application of these techniques between them may also be different.

The differences in the design of e-voting solutions for ranked voting makes analysis between such solutions complicated and difficult. In this paper, we aim to address this problem by proposing a taxonomy of existing e-voting solutions for ranked methods in the literature. To the knowledge of the authors, this is the first time that such a taxonomy has been proposed for ranked methods in particular. Our focus is on the varying properties between e-voting solutions for ranked methods, how these properties are defined, and how these properties influence the decisions which must be made when designing such solutions for ranked methods. We list our contributions as follows.

1. We study the literature for papers which provide e-voting solutions for ranked voting. Our findings demonstrate that the papers may be largely classified according to the amount of information leakage maintained by each solution;

2. We study the cryptographic and architectural designs for the solutions in each of our proposed classes, and explore how the variation in design impacts not only privacy but also additional properties spanning those in both cryptography and social choice;

3. We discuss the implications of our proposed taxonomy and present a number of open problems within the area of secure e-voting for ranked methods.

The remainder of the paper is organised as follows. In Section 2, we introduce ranked voting and introduce definitions for security properties in e-voting. In Section 3, we present our taxonomy of e-voting systems from the literature for ranked voting. In Section 4, we discuss the implications of our proposed taxonomy and identify a number of key open problems relating to the design of e-voting systems for ranked voting. Finally, Section 5 concludes the paper.

2 Background

In this Section, we present introductions to both ranked voting, as it is defined in the literature for social choice theory, and secure e-voting, as it is defined in the cryptographic

literature. We explore the operations and properties associated with popular methods for ranked voting, and define the properties that secure e-voting solutions aim to achieve.

2.1 Ranked Voting

Many different names appear in the literature for "ranked voting", e.g., *preferential voting* [Ka04] and *ranked-choice voting* [Sa21], though we only use the terms "ranked voting" or "ranked methods" in this paper. In a ranked voting system, the voter must take the set C of n choices and list them in order of preference. As an example, suppose a group of voters is trying to decide on the best animal between the choices in $C = \{\texttt{Antelope}, \texttt{Buffalo}, \texttt{Cheetah}, \texttt{Dolphin}\}$. Each voter must construct an ordering ($\succ$) of these choices to cast as their vote, e.g., $\texttt{Dolphin} \succ \texttt{Buffalo} \succ \texttt{Antelope} \succ \texttt{Cheetah}$. In this ordering, the voter prefers `Dolphin` to all other choices, `Buffalo` to `Antelope` and `Cheetah` (but not `Dolphin`), `Antelope` to `Cheetah` (but not `Dolphin` nor `Buffalo`), and `Cheetah` to no other choices.

A voter may also be allowed to submit a partial ordering of the choices as their vote. For example, a voter with a strong opinion between only `Antelope` and `Dolphin` may construct their ordering as simply $\texttt{Dolphin} \succ \texttt{Antelope}$ whilst omitting `Buffalo` and `Cheetah`. Additionally, a voter may express equality or indifference between the choices in their ranking. A voter who cannot decide between `Buffalo` and `Cheetah` for their favourite choice may choose to construct their ordering as $\texttt{Buffalo} \sim \texttt{Cheetah} \succ \texttt{Antelope} \succ \texttt{Dolphin}$. The relation ($\sim$) denotes equality between two or more choices. The voter prefers both `Buffalo` and `Cheetah` equally to `Antelope` and then `Dolphin`.

Once all votes have been cast, the rankings in each vote must be aggregated into a collective tally so that one or more winners can be determined. The tallying process is the biggest differentiator between various ranked voting methods. The most popular ranked methods in the literature include *Borda count*, *Condorcet voting*, *Single Transferable Vote* (STV), and *Instant Runoff Voting* (IRV). We now provide an overview of each of these voting methods.

2.1.1 Borda count

The Borda count voting method assigns a score to each entry in a ranking. The lowest-ranked choice receives a score of 0, the second lowest a score of 1, and so on, up to the highest-ranked choice that receives a score of $n - 1$. For example, the ranking $\texttt{Dolphin} \succ \texttt{Buffalo} \succ \texttt{Antelope} \succ \texttt{Cheetah}$ gives the following mapping of choices to scores.

$$\{\ (\texttt{Antelope}, 1), (\texttt{Buffalo}, 2), (\texttt{Cheetah}, 0), (\texttt{Dolphin}, 3)\ \} \tag{1}$$

The scores for the choices across all rankings are then summed, and the choice with the highest overall score wins the election [Re01]. If a vote contains a partial or equal preference,

then a method is needed to distribute the scores amongst the tied or omitted choices in the vote. Any omitted choice is often assigned a score of 0. Tied choices may be given the minimum score for their position in the ranking, or alternatively may receive half a point for each other choice that they are tied with [NW14]. Borda count has seen usage for national elections in both Nauru and Slovenia [FG14].

2.1.2 Condorcet voting

Condorcet voting interprets rankings differently in comparison to Borda count. A ranking is considered as a series of head-to-head contests between each choice, often represented through a *Pairwise Comparison Matrix* (PCM). For example, the ranking `Dolphin` $\succ$ `Buffalo` $\succ$ `Antelope` $\succ$ `Cheetah` results in the following PCM.

$$\begin{array}{c} \\ \texttt{Antelope} \\ \texttt{Buffalo} \\ \texttt{Cheetah} \\ \texttt{Dolphin} \end{array} \begin{array}{c} \begin{array}{cccc} \texttt{Antelope} & \texttt{Buffalo} & \texttt{Cheetah} & \texttt{Dolphin} \end{array} \\ \begin{pmatrix} - & 0 & 1 & 0 \\ 1 & - & 1 & 0 \\ 0 & 0 & - & 0 \\ 1 & 1 & 1 & - \end{pmatrix} \end{array} \tag{2}$$

In a PCM $\mathbf{M} = (m_{ij})$, $m_{ij} = 1$ and $m_{ji} = 0$ if $i \succ j$ in a ranking. All entries m_{ii} are ignored. At the end of an election, all PCMs are summed into a single matrix $\mathbf{S} = (s_{ij})$. The winner of an election is then taken to be the w such that $s_{wj} > s_{jw}$ for all other $j \in C \setminus \{w\}$. The choice w is called the *Condorcet winner* and is guaranteed to be the socially-optimal choice in an election [Yo88]. Condorcet voting has been utilized for internal decisions by many groups and organisations, including the Wikimedia Foundation, Debiant, Gentoo, Ubuntu, K Desktop Environment (KDE), the Pirate Party of Sweden, OpenStack, ICANN, ACM, IEEE, USENIX, and Google [Sc18].

In an ideal case, a Condorcet winner exists for every election. However, it is possible that the choice w does not exist due to the presence of cycles in the final summed matrix, which may occur even if the PCMs contain no cycles themselves (a *cycle* is the presence of a choice $c \in C$ in a ranking more than once, i.e., the ranking $c \succ \ldots \succ c$). This result is famously known as the *Condorcet paradox* [Ge02]. Various proposals exist in the literature which attempt to extend Condorcet voting and elect an alternative winner in the event that a Condorcet winner does not exist. Following the naming convention of Fishburn [Fi77], these may be referred to as *Condorcet-consistent C2 functions*. Examples include the works of Black [Bl58], Schulze [Sc18], Simpson and Kramer [Kr77; Si69], Copeland [Co51], and Tideman [Ti87]. A digital system which implements any Condorcet-consistent C2 function may be considered an e-voting solution for Condorcet voting.

2.1.3 STV & IRV

Carl Andræ (1812 – 1893) proposed in 1855 a ranked voting system which elects multiple winners. Prior to an election, a desired number of winners is specified, and a *quota* is computed, which determines the tally needed for a choice to be elected a winner. Then, the first preferences across all rankings are summed into a tally, and any choice whose result is at least the quota is elected a winner. This process repeats over multiple rounds, and if the first preference for a voter is already elected, then their vote is said to be a *surplus*, and is instead transferred to count towards the tally for the most-preferred, non-elected choice in their ranking [Ti95].

In 1857, Thomas Hare (1806 – 1891) independently discovered Andræ's proposal. Hare developed a revision in 1865, with a new procedure that eliminates the choice from the election and removes it from all rankings whenever there is no choice that reaches or exceeds the quota. The counting process then continues, electing or eliminating choices as necessary, until all winners have been determined. This forms the basis of the *Single Transferable Vote* (STV) voting method [Ti95]. Many variations exist with differences regarding how the quota is computed, how surplus votes may be selected and transferred, and how ties may be broken whenever choices qualify for election or elimination in a round [Lu06; ON04; Ti95]. Around 1870, a modified version of STV was proposed by William Ware[3], which keeps the elimination process but elects only a single winner. Ware's modification maintained popularity in the literature for social choice theory, and is sometimes referred to as the *Alternative Vote* (AV) (e.g., in the United Kingdom), or as *Preferential Voting* (e.g., in Australia). More commonly however, it is referred to as *Instant Runoff Voting* (IRV). STV and IRV have seen adoption for many local and statewide elections in the United States [TUK22], the national legislature of Australia [Ma03], the local government elections in Scotland [Sc11], and the party leadership elections in the United Kingdom [Pa20b].

2.2 Secure E-voting

In this paper, we take the term "*e-voting*" to refer to the application of any digital system towards the casting or counting procedures in an election. This is the definition proposed by both the Office for Democratic Institutions and Human Rights (ODIHR) and the International Institute for Democracy and Electoral Assistance (IDEA). In particular, it refers to the "*use of electronic voting systems, ballot scanners, and Internet voting*" in an election [VB16].

In general, there are two distinct types of voting, namely, *local* and *remote* voting. Digital technologies may be utilised in either application to create an e-voting system. Local voting is conducted in a defined location, such as a polling station, in a *supervised* environment that allows the voter to immediately ask for help or raise any concerns regarding the election. The casting and tallying processes in local voting can be automated with the assistance

[3] Condorcet also considered such a procedure in 1788, albeit as its own voting method [Na82].

of *Direct Recording Electronic* (DRE) machines: general-purpose devices installed with specific election software [VB16]. On the other hand, remote voting is conducted over no set location in an *unsupervised* environment, e.g., by using a voting server and a web application to allow casting and tallying of votes over the Internet (sometimes referred to as *i-voting*). Remote voting often requires the voter to cast their vote using their own device, such as through their personal computer or mobile phone. However, the introduction of digital technologies to voting systems also introduces significant threats to the secrecy of the votes and the integrity of the election result. It is imperative that an e-voting system maintains two key properties: *verifiability* and *vote privacy*.

2.2.1 Verifiability

Real-world e-voting systems are subject to manipulation. This could be through active attempts by an adversary, or through unintended defects in the installed election software. It is important to protect the integrity of an election; the voters should be able to see that their vote consists of their intended choice and no other, and the final outcome should result from the correct tally of all the votes. The notion of verifiability captures the integrity of an election. Sako and Killian [SK95] initially described the following two variants of verifiability.

- *Individual verifiability*: A voter can verify that their vote is included in the election;
- *Universal verifiability*: Anyone can check that the election outcome is the result of tallying all the votes.

A refined notion of verifiability is *End to End* (E2E) *verifiability* [SH16]. This involves the following verification checks.

- *Cast as intended*: Any voter may verify that their vote truly represents their intended choice and not another;
- *Recorded as cast*: The voter is given information (e.g., a receipt) to verify that their vote has been recorded and included in the tallying process;
- *Tallied as recorded*: Anyone (including a third-party observer) may verify that the final tally is the result of aggregating each recorded vote.

In our taxonomy, we consider individual verifiability to be the composition of “cast as intended” and “recorded as cast”, as these verification checks must be performed by a voter themselves. We then consider universal verifiability to encompass “tallied as recorded”, as this verification check may be performed by any interested party. It is common for e-voting systems to publish information onto a public bulletin board in order to guarantee verifiability

and enable these verification checks to be possible; this may include a shuffle of the votes and the full tally for the election [Hu22].

2.2.2 Vote Privacy

Each vote should only be known to the individual who cast it. This is a widely recognised human right which is described via Article 21 of the Universal Declaration of Human Rights (UDHR) [UN48]. Without vote privacy, voters are subject to intimidation, coercion, and vote buying. Vote privacy is often described through the following security properties [AM16].

- *Ballot secrecy*: A voting system must not reveal the vote for a voter;
- *Receipt freeness*: A voting system should not provide any evidence in the receipt[4] that enables the voter to prove to a third party how they voted;
- *Coercion resistance*: A voter should be able to vote for their intended choice even whilst being influenced by a coercer.

These three properties are the most common descriptors of vote privacy in the literature, with each property implying the previous one, i.e., coercion resistance implies receipt freeness, and receipt freeness implies ballot secrecy [AM16]. However, the satisfaction of these properties alone may not be sufficient to guarantee the privacy of an individual's vote, since private information in a vote may be revealed through the published election result (e.g., consider the extreme case when every voter unanimously chooses the same candidate). This is more likely a concern for e-voting systems that implement ranked voting. In plurality voting, a vote contains limited information as it can only count towards one of a few choices. However, in ranked voting, each vote is an ordering of all or some of the choices in the election and hence some votes are much less likely to be cast than others. This presents the following attack scenario when counts of different rankings are published after the election: an adversary coerces a voter to list a particular choice as their first preference, with the remaining choices listed in an obscure ordering chosen by the adversary themselves. At the end of the election, the coercer must simply check that the vote with the obscure ordering appears on the public bulletin board. This attack is commonly referred to as the *Italian attack* [Be09].

2.2.3 Beyond Vote Privacy

Vote privacy is a crucial property to maintain for an election. However, it does not capture the evolving nature of digital technology. Cryptography that is used to secure elections today

[4] This definition does not require receipts to be avoided entirely; often it is sufficient to obfuscate (e.g., encrypt) the contents of a receipt to hide the plaintext vote.

may not be sufficient to protect the privacy of the votes in many years' time. An adversary may be content with waiting several years for the security of an election to weaken enough so that they can determine through the published data whether a coerced voter followed their instructions. This can be devastating for elections with long-term outcomes such as political or university elections. In order to model the security of elections against a computationally unbounded adversary, researchers have proposed the notion of *everlasting privacy* [Ar13]. E-voting systems which aim for everlasting privacy are not reliant on cryptography that may be weakened at some point in the future, such as computational hardness assumptions.

Furthermore, a key aspect of many e-voting systems in the literature is the reliance on trustworthy tallying authorities (TAs). The TAs must be trusted to perform cryptographic operations and manage their own long-term decryption keys. However, the inclusion of these TAs introduces new concerns regarding the implementation of e-voting systems. The TAs must have the necessary expertise to perform the complex cryptographic operations and manage their own shares of encryption keys. If the TAs lose their key shares, then the election result may become incomputable. The TAs may choose to delegate their tasks to other individuals, however the other individuals must then be trusted as well. Such concerns materialised in practice for Helios [Ad08], an E2E verifiable e-voting system for plurality voting. The chosen TAs understood little about cryptography, and needed to rely upon external experts to perform cryptographic operations. The complexity of managing the TAs has led researchers to build e-voting systems which avoid TAs entirely.

Research on TA-free e-voting started in a decentralized setting, for small-scale boardroom elections. Kiayias and Young proposed the first self-tallying boardroom voting protocol [KY02], followed by the more efficient solutions due to Groth [Gr04] and Hao, Ryan and Zielinski [HRZ10]. Later, Hao et al. [Ha14] extended the self-tallying technique in a decentralized boardroom setting to a centralized setting, and introduced a Self-Enforcing E-Voting (SEEV) paradigm, providing E2E verifiability without involving trustworthy tallying authorities for large-scale elections. Shahandashti and Hao [SH16] proposed an improved SEEV protocol called DRE-ip, which has been used in real-world deployments, e.g., Gateshead polling station voting [Ha20], Durga Puja online voting in New Town, India [Dr24], and UNITA online elections across twelve European universities [Ha25].

3 Proposed Taxonomy

We now present our taxonomy of papers from the literature. For a paper to be included in our taxonomy, it must describe a system with an e-voting component that implements a ranked method from Section 2. Our survey hence identified 38 papers [Ad03; BAH19; Be09; Bl22; CCM08; CGG22; CGY22; Co25; CPP13; Cu15; DTY21; GG05; Gr05; Ha24; HBH24; He07; He21; HPT20; HRT10; Hu22; Jo14; KK08; LD05; LH15; MC05; Pa20a; PB09; Pe09; Ra19; Sm05; Su23; TD24; TPL13; WB09; WL04; Xi07; Xi10; Ya17] which aim to provide e-voting for ranked elections. In our survey, we identified that these solutions

may largely be classified according to their degree of information leakage. We organise these solutions across the following categories.

- **BAL-PUB**: A ranked e-voting system is *ballot public*, i.e., in BAL-PUB, if the link between the voter and their ballot is hidden but the plaintext ballot itself is public;
- **BAL-HID**: A ranked e-voting system is *ballot hidden*, i.e., in BAL-HID, if the plaintext ballots are hidden from the public;
- **TAL-HID**: A ranked e-voting system is *tally hidden*, i.e., in TAL-HID, if the individual plaintext tallies remain hidden and are not published during the election.

Each class represents a differing degree of information leakage. The solutions in BAL-PUB maintain secrecy by hiding the link between the voter and their ballot, but are susceptible to information leakage from the published ballots. The solutions in BAL-HID hide individual ballots and show only the tallies in the final result, e.g., in Condorcet, Borda count, or tallies in each round in a multi-round election, e.g., in IRV or STV. The publication of the tallies allows the public to apply universally agreed electoral rules to determine winners by themselves. In real-world elections, tallies are generally expected by the public and are almost always published. However, tallies may reveal additional information for a ranked election, such as the margin of victory for a Condorcet winner, or the order in which choices are eliminated during an IRV or STV election. Solutions in TAL-HID provide the theoretically strongest guarantee of privacy hiding tallies; only the names of the winners are revealed. These solutions are primarily of academic interest to show theoretical possibilities; they are not used in real-world elections.

We note that our classification of TAL-HID encapsulates the definitions of both publicly tally-hiding e-voting and fully tally-hiding e-voting as described by Huber et al. in their work [Hu22]. In publicly tally-hiding schemes, the tallies are hidden from the public but known to the TAs, whilst in fully tally-hiding schemes, the tallies are not even known to the TAs. Constructing fully tally-hiding schemes is possible by applying additional cryptographic primitives such as multi-party computation (MPC), but the complexity of the resultant system is high. Since the difference between publicly tally-hiding schemes and fully tally-hiding schemes is subtle and of limited relevance to real-world elections, we therefore opt to group these two definitions together in our work. However, we will still consider the impact of TAs on the design of systems within our classification and examine the methods by which some systems aim to avoid TAs entirely. For each paper, we consider the following attributes for comparison.

- *Voting methods*: The supported ranked methods include Borda count, Condorcet voting (with any C2 functions), IRV, and STV;
- *Voting flexibility*: Whether partial and/or equal rankings are permitted;

	Voting Methods					Flexibility		Verifiability			Use of Cryptographic Primitives								Other Properties		
References	*Borda Count*	*Condorcet*	*Condorcet C2*	*IRV*	*STV*	*Partial Ranking*	*Equal Ranking*	*Individual*	*Universal*	*E2E*	*Mix-nets*	*Homomorphism*	*ZKPs*	*Digital Signatures*	*Lattices*	*Pairings*	*Blockchain*	*Code Voting*	*Fake Credentials*	*Free from TAs*	*Everlasting Privacy*	**Local or Remote**
PAV: All-in-One [Xi07, §4]	○	○	○	●	○	●	○	●	●	●	●	●	○	●	○	○	○	○	○	○	○	L
Joaquim [Jo14, §4]	●	●	●	●	●	●	○	●	●	●	●	○	●	○	○	○	○	●	○	○	○	L
Aditya et al. [Ad03]	●	●	●	●	●	○	○	○	●	○	●	○	○	○	○	○	○	○	○	○	○	L
Belenios [CGG19; CGG22; Co23; Gl25]	○	●	◐	●	●	●	●	●	●	●	●	○	●	●	○	○	○	○	○	○	●	R
vVote [Cu15]	○	○	○	●	●	○	○	●	●	●	●	○	○	●	○	○	○	○	○	○	○	L
Heather et al. [HRT10]	●	○	○	●	●	○	○	○	●	○	●	●	●	○	○	○	○	●	○	○	○	R
Zeus [TPL13]	○	○	○	●	●	●	○	●	●	●	●	○	●	●	○	○	○	○	○	○	○	R
Civitas [CCM08]	○	●	●	○	○	○	●	○	●	○	●	●	●	●	○	○	○	○	●	○	○	R
Peng [Pe09]	●	●	●	●	●	○	○	○	●	○	●	●	●	○	○	○	○	○	○	○	○	L
Vote & Check [Co25]	●	●	●	●	●	○	○	●	●	●	○	○	○	●	○	○	○	○	○	○	●	R
Locher and Haenni [LH15]	●	●	●	●	●	○	○	●	●	●	○	○	●	○	○	○	○	○	○	○	●	R
Cuvelier et al. [CPP13]	●	●	●	●	●	○	○	○	●	○	●	○	●	○	○	○	○	○	○	○	●	L

● – a fulfilled attribute; ◐ – a partially supported attribute or support is only briefly considered; ○ – an attribute not fulfilled.

Tab. 1: The BAL-PUB categorisation of ranked e-voting systems. Note that "Condorcet C2" refers to any Condorcet-consistent C2 function. We follow what is primarily described in the cited work regarding its setting (local or remote).

- *Verifiability*: Individual, universal, and/or E2E verifiability;
- *Cryptography*: The cryptographic techniques utilised by the solution;
- *Additional Properties*: Relevant properties include everlasting privacy and any need for TAs;
- *Setting*: Whether the proposed solution is primarily described for local or remote voting in the original paper.

We now provide an analysis of each category in our taxonomy.

3.1 BAL-PUB

First, we consider the BAL-PUB category of ranked e-voting systems. This category covers 12 schemes [Ad03; CCM08; CGG22; Co25; CPP13; Cu15; HRT10; Jo14; LH15; Pe09; TPL13; Xi07]. An overview of our analysis is presented in Table 1.

3.1.1 Cryptographic Primitives

Almost all of the papers in this category rely on *mix-nets* to provide ranked e-voting. A *mix* receives an ordering of ciphertexts, removes a layer of encryption, shuffles the items, and returns the result. A mix-net is multiple mixes connected together [AM16], and guarantees privacy by mimicking the traditional paper ballot box. Each ballot is shuffled with every other ballot in the mix-net, making it impossible for any party other than the TAs running the mix-net to trace a vote back to the individual that cast it. After the final mix, the votes

are revealed in plaintext through a permuted order on the public bulletin board and can be tallied by any observer. The need to reveal all votes in plaintext makes such systems vulnerable to the Italian attack, hence the placement of these systems into BAL-PUB. We note that two works[5] [Jo14; Xi07] discuss multiple approaches which could be taken when tallying the votes in a ranked election; here, we place those approaches that rely primarily on mix-based tallying into BAL-PUB (later, in Section 3.2, we consider the alternative approaches offered by these papers in tallying ranked ballots).

One work by Peng and Bao [PB09] opts to divide the ballots into groups and tally the groups by exploiting *homomorphic encryption*. Homomorphic encryption allows for the composition of encrypted messages, without knowledge of any secret key, in order to create a new ciphertext containing either the sum or the product of the underlying messages [Ha23]. A homomorphic encryption scheme is *additive* if the composition of encrypted messages yields an encryption of the sum of both messages, and it is *multiplicative* if this composition instead yields the encryption of the product of both messages. In an e-voting system, ballots are typically encrypted using additive homomorphism and composed into a single ciphertext containing the sum of all votes. The TAs then decrypt the result to reveal the final tally. The work of Peng and Bao [PB09] adopts a hybrid approach combining mix-nets with multiplicative homomorphism. Votes are represented as prime numbers which are multiplied together during the tallying process and later factorized to reveal the plaintext votes. The threat of the Italian attack remains due to the factorization revealing each vote at the end of an election.

Some solutions in BAL-PUB extend existing e-voting systems for plurality voting to support ranked methods [CCM08; Cu15; HRT10; TPL13; Xi07]. Two works extend Prêt à Voter [Ry09], with the work of Xia et al. [Xi07] designed primarily for IRV elections and the work of Culnane et al. [Cu15] designed for full STV elections. Other e-voting systems used as a basis for solutions in BAL-PUB include Helios [Ad08], Pretty Good Democracy [RT09], and the e-voting scheme developed by Juels, Catalano, and Jakobsson [JCJ05]. The latter scheme is refined by Clarkson et al. [CCM08] to permit Condorcet voting whilst allowing voters to fake their own credentials when casting their vote, which helps to provide coercion resistance. However, this does not eliminate the possibility of coercion as a result of information leakage in later stages of an election such as during the tallying process.

A majority of the solutions in BAL-PUB utilise *Zero Knowledge Proof* (ZKP) techniques. ZKPs, first proposed by Goldwasser, Micali, and Rackoff [GMR89], enable a party to prove that a statement is true without revealing any information other than the correctness of that statement. In e-voting, ZKPs are often adopted to allow the system to prove that a ballot is *well-formed*, i.e., it encrypts one of the possible rankings for an election, without revealing any information about the underlying plaintext vote. This ensures that a voter does not submit a vote for a choice not running in the election, or submit a vote that contains the

[5] The work of Aditya et al. [Ad03] also considers multiple approaches for ranked e-voting, however only their mix-based approach [Ad03, §3] covers ranked methods. The authors also consider a design using homomorphic encryption [Ad03, §2], however it is not clear which ranked methods are covered by this design.

same choice multiple times. Furthermore, we see that some solutions [CCM08; CGG19; Co25; Cu15; TPL13; Xi07] adopt *digital signature* schemes. The use of digital signatures in these solutions ensures the authenticity of any generated receipt and the content it provides, including the authenticity of encrypted ballots and associated ZKPs.

3.1.2 Ballot Construction

In general, one of three approaches is taken in order to encrypt ranked ballots within BAL-PUB. The first is to assign each ranked ballot to an identifier which is then encrypted [Ad03; Pe09; TPL13]. The second is to represent a plaintext vote as a matrix, which could be either a PCM for Condorcet voting or a permutation matrix for other ranked voting methods [CCM08; HRT10]. The third is to directly encrypt a ranking represented as a permutation into a single ballot [He07]. The identifier-based approach requires reconstruction of the original ballots at the end of the election, which incurs $O(n!)$ additional work. The latter two approaches avoid this reconstruction step but require $O(n^2)$ and $O(n)$ additional storage respectively. We note that ranked e-voting systems may utilise different methods of ballot representation as necessary; the work of Heather et al. [HRT10] presents multiple protocols which vary between ballots using encrypted matrices or encrypted rankings. This creates a trade-off between the complexity of vote casting and verification in their work [HRT10]. Finally, we note that the works of Heather et al. [HRT10] and Joaquim [Jo14] support *code voting*. In code voting, each voter receives a code sheet containing voting codes, one for each choice in the election. The voter replaces the choices in their ranking by substituting the corresponding voting codes instead. This aims to provide protection against an untrusted voting device that attempts to learn a voter's ranking.

3.1.3 Voting Methods and Flexibility

IRV and STV are the two most common voting methods supported by systems in BAL-PUB. Around half of the solutions in BAL-PUB support Borda count or Condorcet voting. Of the solutions that support Condorcet voting, all of them also support various Condorcet-consistent C2 functions. These solutions are not explicitly designed with this support in mind; rather, this is a benefit of their tallying processes. All votes are revealed in plaintext at the end of an election, allowing straightforward tallying by direct application of any Condorcet-consistent C2 function over the plaintext votes.

Some works [CGG19; Jo14; TPL13; Xi07] allow for the voter to cast a partial ranking as their vote. A novel method to permit partial rankings in ranked e-voting systems is to include an additional choice, often named "stop", which acts as a cut-off point in a ranking. The first portion of the ranking preceding "stop" is taken as the vote, whilst the latter portion beyond "stop" contains the remaining choices in an arbitrary order [He07]. This approach transforms a partial ranking into a complete one, enabling these two kinds of ranking to be

handled much in the same way in an e-voting system. Xia et al. [Xi07] utilise this approach in their work. Tsoukalas et al. [TPL13] introduce additional identifiers for partial rankings which are encrypted into ballots, however this approach is less scalable to large elections. Joaquim [Jo14] opts to pad a partial ranking with dummy entries so that the size of a partial ranking and a complete one are the same prior to encryption, however it is then more complex to check if a ballot is well-formed compared to a solution using a "stop" choice. Only two works [CCM08; CGG19] permit the voters to express equality or indifference between the choices in their ranking.

3.1.4 Other Security Properties

We find that all solutions in BAL-PUB achieve universal verifiability, with most additionally achieving individual verifiability and therefore E2E verifiability. The main difference between these solutions that achieve E2E verifiability and those that do not is that the former set of solutions ensure votes are cast as intended [CPP13; Cu15; TPL13; Xi07]. The voter is provided a means to audit their cast vote. Voter-initiated auditing, proposed by Chaum [Ch04] and Benaloh [Be06], allows a voter to check the accuracy of the vote casting procedure by performing the encryption themselves. Audited votes are never included in the final tally, and the voter may choose to audit their ballot any number of times. The voter's choice is recorded in the election for tallying only when the voter chooses to confirm the ballot.

We observe that few solutions in BAL-PUB are designed with everlasting privacy in mind [CGG22; Co25; CPP13; LH15]. The techniques to achieve everlasting privacy differ between these solutions. One work ensures no correspondence between a voter and their cast ballot but only when revoting is disallowed [CGG19]. Another work associates each voter with a pseudonym that is published alongside their ballot on the bulletin board, although the link between the voter and their pseudonym must remain secret [Co25]. The remaining works realise everlasting privacy by relying upon perfectly hiding commitment schemes [CPP13; LH15]. Under such schemes, it should be infeasible to link a recovered plaintext vote to the individual who cast it, however this does not prevent information leakage which could arise from the contents of decrypted ballots. Finally, we note that no solution in BAL-PUB is designed without reliance on any TAs.

3.1.5 Summary

The solutions in BAL-PUB primarily utilise mix-based tallying for ranked elections, which provides only a weak guarantee of privacy due to the threat of the Italian attack. This is a significant threat for ranked e-voting systems including those which strive for everlasting privacy. All four solutions [CGG22; Co25; CPP13; LH15] considered thus far that satisfy everlasting privacy are susceptible to the Italian attack. Most systems in BAL-PUB achieve

	Voting Methods					Flexibility		Verifiability			Use of Cryptographic Primitives									Other Properties		
References	*Borda Count*	*Condorcet*	*Condorcet C2*	*IRV*	*STV*	*Partial Ranking*	*Equal Ranking*	*Individual*	*Universal*	*E2E*	*Mix-nets*	*Homomorphism*	*ZKPs*	*Digital Signatures*	*Lattices*	*Pairings*	*Blockchain*	*Code Voting*	*Fake Credentials*	*Free from TAs*	*Everlasting Privacy*	**Local or Remote**
PAV: All-in-One [Xi07, §5]	●	●	○	○	○	●	○	●	●	●	○	●	○	●	○	○	○	○	○	○	○	L
Joaquim [Jo14, §5]	●	○	○	○	○	●	○	●	●	●	○	●	●	○	○	○	○	●	○	○	○	L
Xia et al. [Xi10, §4.6]	○	●	○	○	○	○	○	●	●	●	●	●	●	●	○	○	○	○	○	○	○	L
Ramchen et al. [Ra19]	○	○	○	●	○	●	○	○	●	○	○	●	●	○	○	●	○	○	○	○	○	L
Goh and Golle [GG05]	○	○	○	●	○	○	○	○	●	○	○	●	●	○	○	○	○	○	○	○	○	L
Heather [He07]	○	○	○	●	◐	●	○	●	●	●	●	○	●	●	○	○	○	○	○	○	○	L
Myers and Clarkson [MC05]	○	●	○	○	○	○	○	●	●	●	●	●	●	●	○	○	○	○	●	○	○	R
Keller and Kilian [KK08]	○	○	○	●	○	◐	○	○	●	○	●	●	●	○	○	○	○	○	○	○	○	L
Black et al. [Bl22]	●	○	○	●	○	◐	○	○	●	○	○	●	●	●	●	○	○	○	○	○	○	L
Haines et al. [HPT20]	○	●	◐	○	○	○	●	○	●	○	●	●	●	○	○	○	○	○	○	○	○	L
Smith [Sm05]	○	○	○	●	●	○	○	○	●	○	●	●	●	○	○	○	○	○	●	○	○	L
Yang et al. [Ya17]	○	○	○	○	○	○	○	○	●	○	○	●	●	●	○	○	○	○	○	○	○	R
Groth [Gr05]	●	○	○	○	○	○	○	○	○	○	○	●	●	○	○	○	○	○	○	○	○	L
Sun et al. [Su23]	●	●	○	○	○	○	○	○	●	○	○	○	○	○	○	○	●	○	○	○	○	R
Panja et al. [Pa20a]	●	○	○	○	○	○	○	●	●	●	○	●	●	●	○	○	●	○	○	●	○	R
VERICONDOR [Ha24]	○	●	◐	○	○	●	●	●	●	●	○	●	●	●	○	○	○	○	○	●	○	L
Camel [HBH24]	○	○	○	●	○	○	○	●	●	●	○	●	●	●	○	○	○	○	○	●	○	L
DRE-Borda [BAH19]	●	○	○	○	○	○	○	●	●	●	○	●	●	●	○	○	○	○	○	●	○	L

● – a fulfilled attribute; ◐ – a partially supported attribute or support is only briefly considered; ○ – an attribute not fulfilled.

Tab. 2: The BAL-HID categorisation of ranked e-voting systems. Note that "Condorcet C2" refers to any Condorcet-consistent C2 function. We follow what is primarily described in the cited work for its setting (local or remote).

E2E verifiability, though all require TAs to run mixing or decryption operations. The main benefit of the solutions in BAL-PUB is the ease with which various ranked methods can be supported, as all tallying can be performed in plaintext at the end of an election.

3.2 BAL-HID

We now review the BAL-HID category of ranked e-voting systems, which includes 18 solutions [BAH19; Bl22; GG05; Gr05; Ha24; HBH24; He07; HPT20; Jo14; KK08; MC05; Pa20a; Ra19; Sm05; Su23; Xi07; Xi10; Ya17]. We present an overview of our analysis in Table 2.

3.2.1 Cryptographic Primitives

Almost all solutions in BAL-HID rely on homomorphic encryption to provide security for ranked e-voting. Most of these systems [BAH19; GG05; Gr05; Ha24; HBH24; HPT20; Jo14; Pa20a; Ra19; Sm05; Xi07; Xi10; Ya17] exploit specifically additive homomorphism for tallying ballots; this is in contrast to the solutions in BAL-PUB which rely primarily on mix-based tallying. Additively homomorphic tallying enables the aggregation of ballots into a single encrypted tally that, once decrypted, can be directly inspected to reveal the election result. Only the final tally is revealed, which mitigates concerns regarding the Italian attack.

Some solutions utilise additional cryptographic techniques, whilst others opt for different tallying approaches entirely. Two works apply mix-nets in the tallying of ranked ballots,

however the application is different in comparison to most of the solutions in BAL-PUB [He07; KK08]. Both of these works handle either IRV or STV elections, and perform the tallying process by decrypting only the most-preferred choice in each ballot during a round. The whole vote is never visible all at once; this is in contrast to the solutions in BAL-PUB that reveal the whole vote when a ballot is decrypted as a result of mixing. The votes are then re-encrypted or re-mixed for later rounds [He07; KK08]. Black et al. [Bl22] describe an extension to the EVOLVE [De17] e-voting system and provide new tallying processes for Borda count and IRV elections. EVOLVE [De17] is a lattice-based e-voting scheme designed primarily with post-quantum security in mind. As part of their scheme, each voter commits to their vote and provides a Hamming weight ZKP to prove a ballot is well-formed. The commitments are batched to several TAs who then decrypt and aggregate the votes themselves. Anyone can then combine the aggregated batches into the final election result. We observe two works that operate over a blockchain [Pa20a; Su23]. The first requires each voter to mask their ballot before committing to it using a Quantum Bit Commitment (QBC) protocol; the masked ballots are distributed to every miner on the blockchain so that consensus on the election result can be achieved [Su23]. The second is designed primarily for Borda count [Pa20a]. Each voter must commit to their vote on the Ethereum blockchain. A deposit must be paid prior to voting, which is later refunded once a vote has been cast; this acts as an incentive for the voters to engage with the election and not abstain from voting once registered.

Almost all solutions in BAL-HID utilise ZKPs in order to prove a ballot is well-formed and around half utilise digital signatures to provide authenticity of generated ballots and receipts. We observe that some solutions utilise mix-nets in a unique way outside of tallying ballots. The work of Smith [Sm05] adopts mix-nets for the elimination process in STV elections. Transfer of votes from winning choices is done by decrypting the first row of a permutation matrix, rotating it to the end of the matrix, then re-encrypting and mixing the result. The work of Haines et al. [HPT20] considers each ranked ballot to be well-formed if it is a permutation of a known ranked ballot. A mix-net is used to permute the known ballot with the mix bound to the ballot and the permutation hidden through corresponding ZKPs.

3.2.2 Ballot Construction

Ballots are generally constructed in the same manner as those in BAL-PUB, albeit with slight differences. We see solutions that adopt matrix-based representations [Bl22; Ha24; HBH24; HPT20; MC05; Ra19; Sm05; Su23; Xi10; Ya17] as well as those that directly use permutations to represent ranked ballots [BAH19; GG05; Gr05; He07; Jo14; Pa20a; Xi07]. A matrix-based representation is popular for the solutions that utilise homomorphic tallying [BAH19; Gr05; Ha24; HBH24; HPT20; Pa20a; Ra19; Sm05; Xi10; Ya17], which could be either a matrix of pairwise comparisons or a permutation matrix. Representing ballots as matrices is a natural choice for securing Condorcet, IRV, or STV elections as the

corresponding tallying processes can be achieved by simple matrix addition which pairs well with additively homomorphic encryption.

A couple of solutions in BAL-HID adopt alternate representations to those that we have seen thus far. Keller and Kilian [KK08] represent ranked ballots as linked lists for elections in order to facilitate stronger privacy guarantees than solutions in BAL-PUB whilst maintaining only a quadratic amount of computation during ballot construction. Black et al. [Bl22] represent ranked ballots as trees of possible combinations that can be created following the elimination of a choice in an IRV election. Authorities are responsible for updating ballots by choosing the path with the correct combinations as choices are eliminated in an election. However, ballot construction is exponential in this scheme.

Some solutions in BAL-HID must encode additional information into ballots in order to prevent information leakage. This is of particular importance for STV elections. During an STV election, not all votes may contribute to the tallies equally. Often, each vote starts with a unit weight which may be reduced as an election progresses. This reduction corresponds to the method of surplus transfer chosen for the election. For example, the Gregory method of surplus transfer in STV may produce votes with fractional weights [Ti95]. Any e-voting system for STV must take into account how these weights should be represented and tallied. If the weight is public, then an adversary could examine how the weight is updated during tallying in order to glean information about the ranking created by the voter when casting their ballot.

Two e-voting systems support STV elections in BAL-HID [He07; Sm05]. The work of Heather [He07] only provides a complete solution for IRV, but discusses how their solution can be modified to support full STV elections. The work of Smith [Sm05] provides a complete description of an e-voting system for STV. Both solutions discuss the same method for representing weights: an initial value of a ballot is multiplied by a predetermined scaling factor prior to any tallying. When a ballot must be reweighted, any divisions are taken to be integer values rather than fractional ones. Then, any published tallies during the election are scaled down using the scaling factor. This method essentially has each voter cast multiple votes (in amount equal to the scaling factor) rather than a single vote in order to simplify tallying of ballots involving fractional values. The major advantage of this approach is its simplicity: the scaling factor can be public and applied to ballots through homomorphism without revealing any information about the contents of the ballot including its underlying plaintext ranking or its weight. However, the choice of scaling factor introduces new difficulties in the management of STV elections. If the scaling factor is too small, then rounding errors could be large and impact the outcome of an election. Conversely, a scaling factor which is too large would result in the replication of many ballots and long delays in tallying. Smith [Sm05] advises for a scaling factor of 120 (but without justification). The scaling factor must be appropriately balanced for e-voting systems that adopt this method of reweighting ballots.

3.2.3 Voting Methods and Flexibility

The e-voting systems in BAL-HID are much more specialised to particular ranked methods in comparison to the systems in BAL-PUB. No solution in BAL-HID supports more than a couple of ranked methods. This is largely due to the widespread usage of homomorphic tallying as opposed to mix-based tallying in BAL-HID. In mix-based tallying, it is straightforward to support many ranked methods as tallying is performed over plaintext ballots. It is trivial to transform such ballots into PCMs for Condorcet voting or eliminate choices as necessary for IRV and STV elections. Homomorphic tallying must be designed with a specific voting method in mind, as the tallying process for one ranked method is often not applicable to another.

Borda count and IRV are the two most supported ranked methods in BAL-HID. STV is the least supported and covered by only two works [He07; Sm05]. Some solutions support Condorcet voting, although only two of these solutions support specific Condorcet-consistent C2 functions. Haines et al. [HPT20] design an e-voting solution for Condorcet voting that permits Schulze's method [Sc18], whilst Harrison et al. [Ha24], in addition to Schulze's method, design an e-voting solution for Condorcet voting that permits tallying using the works of Black [Bl58], Simpson and Kramer [Kr77; Si69], Copeland [Co51], and Tideman [Ti87]. However, these two e-voting solutions do not provide full support for their implemented C2 functions. Haines et al. [HPT20] do not consider how ties should be broken when two or more choices are found to be winners under Schulze's method. Harrison et al. [Ha24] do not provide full support for tallying using either Schulze's or Tideman's method. Under certain conditions, these two C2 functions require access to plaintext votes in order to break ties [Sc18; Ti87]. The work of Harrison et al. [Ha24] uses homomorphic tallying to aggregate ballots encrypted using ephemeral keys that are later discarded alongside the plaintext votes. This approach was first adopted within the DRE-ip [SH16] protocol to provide self-enforcing plurality voting. As such, the plaintext votes are unavailable by the time they may be needed for tallying under Schulze's or Tideman's method.

Many solutions in BAL-HID adopt similar methods to allow partial rankings as those belonging to BAL-PUB, with the most common being the usage of a "stop" choice [HBH24; He07; Ra19]. The work of Keller and Kilian [KK08] directly encrypts partial rankings without adding a "stop" choice and is susceptible to leaking information regarding the number of choices in a ballot; the authors acknowledge that padding the rankings is a potential solution to this information leakage. Black et al. [Bl22] represent partial rankings by having the voter specify which choices are not included in their ranking when constructing their ballot. However, this approach is only applied to their solution for Borda count. Their solution for IRV does not support partial rankings. Only two works in BAL-HID permit voters to rank choices equally [Ha24; HPT20]. The work of Harrison et al. [Ha24] relies upon submitting any equal rankings as part of a separate voting matrix, whereas the work of Haines et al. [HPT20] constructs equal rankings using the previously noted mix-based

approach that permutes a known ranked ballot. These two solutions only target Condorcet voting, and the former incurs complex ZKPs whilst the latter requires management of mix servers.

3.2.4 Other Security Properties

Almost all solutions in BAL-HID achieve universal verifiability, with some achieving individual (and hence E2E) verifiability. Similar to the solutions in BAL-PUB, the most common method for ensuring votes are cast as intended is to allow for voter-initiated auditing [Ha24; HBH24; He07; MC05; Xi07; Xi10]. Some solutions in BAL-HID aim to provide strong guarantees on privacy and verifiability whilst simplifying election management through the removal of either trustworthy TAs or the need for voters to perform cryptography themselves [BAH19; Ha24; HBH24; Pa20a]. Such solutions have been proposed for Borda count, Condorcet voting (including some support for various C2 functions), and IRV; however, no solution has been proposed thus far for STV elections. Additionally, no solution in BAL-HID is designed for everlasting privacy.

3.2.5 Summary

The solutions examined in BAL-HID largely avoid mix-based tallying in favour of homomorphic tallying to provide stronger guarantees of privacy and mitigate the Italian attack. However, the e-voting solutions in BAL-HID are more specialised to particular ranked voting methods compared to the solutions in BAL-PUB. Few e-voting solutions in BAL-HID support STV elections or Condorcet-consistent C2 functions [Ha24; HPT20], and of those that do provide support, this support is either partial, only briefly discussed, or raises concerns regarding its scalability to large elections. Around half of all solutions in BAL-HID maintain E2E verifiability [BAH19; Ha24; HBH24; He07; Jo14; MC05; Pa20a; Xi07; Xi10], though of these solutions, only few aim to provide self-enforcing e-voting [BAH19; Ha24; HBH24; Pa20a] and no solution is designed to maintain everlasting privacy.

3.3 TAL-HID

We now consider the 11 solutions [Be09; CGY22; DTY21; He21; Hu22; LD05; PB09; TD24; WB09; WL04; Xi10] belonging to TAL-HID. We present an overview of our analysis in Table 3.

	Voting Methods					Flexibility		Verifiability			Use of Cryptographic Primitives									Other Properties		
References	*Borda Count*	*Condorcet*	*Condorcet C2*	*IRV*	*STV*	*Partial Ranking*	*Equal Ranking*	*Individual*	*Universal*	*E2E*	*Mix-nets*	*Homomorphism*	*ZKPs*	*Digital Signatures*	*Lattices*	*Pairings*	*Blockchain*	*Code Voting*	*Fake Credentials*	*Free from TAs*	*Everlasting Privacy*	**Local or Remote**
Xia et al. [Xi10, §4.5]	○	○	○	●	●	●	○	●	●	●	●	●	●	●	○	○	○	○	○	○	○	L
Cortier et al. [CGY22]	○	●	◐	●	●	●	●	○	●	○	●	●	●	●	○	○	○	○	○	○	○	L
Hertel et al. [He21]	●	●	◐	●	○	○	○	●	●	●	●	●	●	●	○	○	○	○	○	○	○	R
Kryvos [Hu22]	●	●	○	●	○	●	○	○	●	○	○	●	●	○	○	●	○	○	○	○	○	L
Shuffle Sum [Be09]	○	○	○	●	●	◐	○	○	●	○	●	●	●	○	○	○	○	○	○	○	○	L
Lee and Doi [LD05]	○	●	○	○	○	○	○	○	●	○	○	●	●	○	○	○	○	○	○	○	○	L
Tassa and Dery [TD24]	●	●	◐	○	○	○	○	○	●	○	○	○	○	●	○	○	○	○	○	○	○	L
Peng and Bao [PB09]	○	○	○	●	○	○	○	○	●	○	○	●	●	○	○	○	○	○	○	○	○	L
Wen and Buckland [WB09]	○	○	○	●	○	●	○	○	●	○	●	●	●	○	○	○	○	○	○	○	○	R
Dery et al. [DTY21]	●	○	○	○	○	○	○	○	●	○	○	○	○	●	○	○	○	○	○	○	○	R
Wang and Leung [WL04]	●	○	○	○	○	○	○	●	●	●	●	●	●	●	○	○	○	○	○	○	○	L

● – a fulfilled attribute; ◐ – a partially supported attribute or support is only briefly considered; ○ – an attribute not fulfilled.

Tab. 3: The TAL-HID categorisation of ranked e-voting systems. Note that "Condorcet C2" refers to any Condorcet-consistent C2 function. We follow what is primarily described in the cited work for its setting (local or remote).

3.3.1 Cryptographic Primitives

Similar to BAL-HID, we see that most solutions in TAL-HID rely upon homomorphic encryption to provide security in their designs. This is combined with additional cryptographic techniques to create fully tally hiding protocols. Several of these solutions are designed as secure multi-party computation (MPC) protocols [CGY22; DTY21; He21; TD24; WL04]. MPC protocols allow a set of trustees to perform any function over private inputs so that every participant learns the result of the function but nothing else [DTY21]. This design approach has a natural application towards e-voting systems aiming to prevent any kind of information leakage that may occur from the election result. Some systems combine MPC with secret sharing [DTY21; He21; TD24]; shares of the votes are distributed between multiple TAs so that no TA alone is trusted for secrecy of the ballots. Some systems instead use range proofs to calculate the election result without revealing any information other than the final winner [Be09; PB09].

We see continued usage of mix-nets in TAL-HID for proving that a ranked ballot is well-formed [CGY22] as well as for tallying the ballots. The Shuffle Sum protocol utilises mix-nets to convert between different ballot formats in order to simplify tallying for STV elections. Four solutions utilise the *mix and match* technique [JJ00] as part of their tallying processes [LD05; WB09; WL04; Xi10]. This technique mixes a tabular representation of two ciphertexts and performs either a Plaintext Equality Test (PET) or a Plaintext Inequality Test (PIT) to allow the contents of both ciphertexts to be compared. The authorities performing the comparison gain no information about the underlying plaintexts. This technique is used in TAL-HID for revealing only the winner for Borda count [WL04], Condorcet [LD05], and IRV [WB09; Xi10] elections. These solutions demonstrate the versatility of mix-nets for ranked elections, although consideration should be taken regarding their usage so as to avoid the Italian attack.

3.3.2 Ballot Construction

Many solutions in TAL-HID adopt similar methods for ballot construction compared to the solutions in BAL-HID or BAL-PUB. Some solutions adopt multiple construction methods for ballots and vary their usage depending on the chosen voting method for an election [CGY22; He21; Hu22]. This can help to optimise the performance of ballot construction for specific voting methods. Cortier et al. [CGY22] utilise encrypted PCMs for Condorcet ballots but vectors of bit-wise encrypted integers for STV ballots. Hertel et al. [He21] also use encrypted PCMs for Condorcet ballots but allow for vectors of encrypted integers to be used for ballots under Borda count, although IRV ballots in their scheme are represented as encrypted identifiers. A caveat of adopting multiple approaches for ballot construction is that additional proofs must be designed to verify that a ballot is well-formed.

Other solutions adopt multiple ballot formats within the same voting method. Benaloh et al. [Be09] use this approach within the Shuffle Sum protocol to improve the performance of tallying for STV elections. Ballots within their scheme vary between a list of encrypted choices, a list of encrypted integers representing the rankings of these choices, a list of encrypted bits flagging the eliminated choices, and a list of encrypted weights such that the most-preferred choice has the correct weight with all other weights equal to zero [Be09]. Ballot weights are updated by multiplying the weight of every elected choice by a and all other weights by d where a/d is the transfer value for an elected winner. The product of all denominators d must be tracked in order to update the quota in later rounds [Be09]. Using this scheme to handle weights in STV elections avoids any issues with precision or large quantities of votes. Cortier et al. [CGY22] approximate fractional values in their scheme for STV elections, whereas Xia et al. [Xi10] adopt the Shuffle Sum protocol directly as part of their scheme for tallying STV ballots.

3.3.3 Voting Methods and Flexibility

Similar to BAL-HID, we observe that the solutions in TAL-HID are largely specialised to cover one or few ranked methods. Only the works of Cortier et al. [CGY22] and Hertel et al. [He21] aim to cover a range of ranked methods, and achieve this by not relying solely upon homomorphic tallying, but by building up the tallying processes through a set of defined MPC building blocks. We note that the work of Cortier et al. [CGY22] does not cover Borda count, whilst the work of Hertel et al. [He21] does not support STV. STV is tied for the least-supported voting method in BAL-HID alongside Condorcet-consistent C2 functions. Only the work of Hertel et al. [He21] provides a comprehensive implementation of various Condorcet-consistent C2 functions as part of their solution, whereas the work of Cortier et al. [CGY22] is limited to the C2 functions proposed by Schulze [Sc18] and Tideman [Ti87]. However, it is unclear how Hertel et al. [He21] manage the breaking of ties that may result from Schulze's method for tallying Condorcet ballots. Tassa and Dery [TD24] also provide some support for Condorcet-consistent C2 functions in their

work, although this support is limited to the C2 functions proposed by Copeland [Co51] and Simpson and Kramer [Kr77; Si69].

Few solutions in TAL-HID support partial rankings. Most utilise a "stop" choice to provide this support [CGY22; WB09; Xi10]. We note that the work of Benaloh et al. [Be09] only provides a brief consideration for how a "stop" choice may be incorporated into the Shuffle Sum protocol without further description for any necessary changes or extensions to their work. Another work assigns each partial ranking to its own identifier that is later encrypted to form a ballot [Hu22]. We observe that only one solution is able to provide support for equal rankings [CGY22]; this is achieved for Condorcet voting by applying an iterative algorithm that transforms a PCM into one that specifies indifference between multiple choices. The resulting matrix is then shuffled with a mix-net [CGY22].

3.3.4 Other Security Properties

Few solutions in TAL-HID achieve E2E verifiability. Huber et al. aim to prove verifiability of the Kryvos protocol under the model of Küsters et al. [KTV10] rather than through the more common descriptors of individual, universal, and E2E verifiability [Hu22]. However, this verifiability model does not capture the "cast as intended" check for the voters and therefore does not completely provide individual verifiability. The work of Tassa and Dery [TD24] is designed under the same model of verifiability but without associated proofs. No solution in TAL-HID is designed with everlasting privacy in mind or designed without reliance on trustworthy TAs.

3.3.5 Summary

The solutions in TAL-HID largely combine homomorphic tallying with additional MPC components to create e-voting systems that are resilient against the Italian attack and prevent information leakage from the published election results. Most of the solutions in TAL-HID are specialised to particular ranked voting methods. There is no comprehensive solution that covers all of the ranked methods we consider in this paper. No solution in TAL-HID completely covers Condorcet elections with various Condorcet-consistent C2 functions. Additionally, no solution is designed to support everlasting privacy or self-enforcing e-voting.

4 Discussion

Figure 1 presents an overview of the research development in ranked e-voting. Each citation in Figure 1 corresponds to the initial proposal of an e-voting primitive or the subsequent application of such a primitive towards ranked elections. The first proposals for secure e-voting include plurality systems designed using either mix-nets [Ch81] or homomorphic

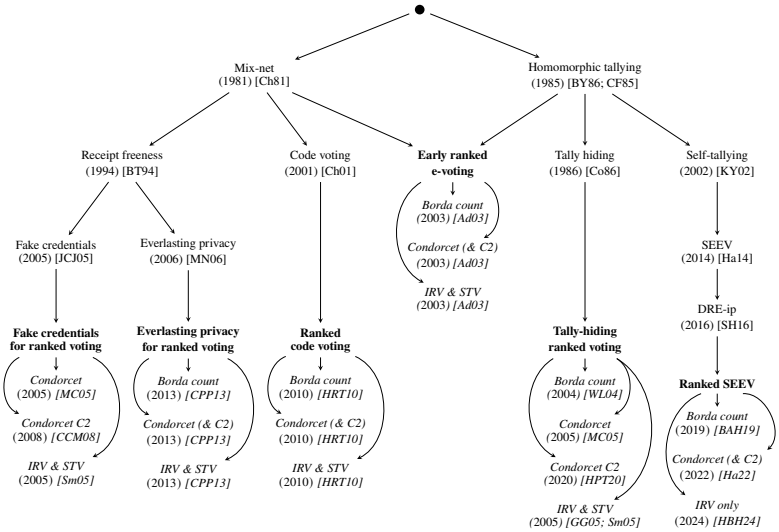


Fig. 1: Research development in ranked e-voting.

encryption [BY86; CF85]. Extensions to these designs produced early e-voting systems for ranked methods [Ad03]. Tally-hiding e-voting was considered for plurality systems as early as 1986 [Co86], with the work of Wen and Buckland [WB09] providing the first design of a tally-hiding system for a ranked election. Receipt-free e-voting systems [BT94] extended early mix-net designs and were eventually refined into e-voting systems that utilise fake credentials [JCJ05] or maintain everlasting privacy [MN06]. Fake credentials were quickly adopted as part of designs for ranking-based e-voting systems [MC05; Sm05]. It was not until 2013 when we saw the first e-voting systems that could handle ranked elections whilst maintaining everlasting privacy [CPP13]. Proposals for code voting also followed early mix-based designs [Ch01], with Heather et al. [HRT10] providing a code voting solution that supports ranked elections. Self-enforcing e-voting (SEEV) refers to E2E verifiable e-voting without TAs; it descends from the homomorphic branch of e-voting systems, and follows research into self-tallying e-voting [KY02]. The first SEEV protocol for plurality voting, named DRE-i, was proposed in 2014 [Ha14]. DRE-i was later refined into DRE-ip in 2016 [SH16]. Following DRE-ip, the work of Bag et al. [BAH19] on DRE-Borda marked the first proposal of a SEEV system for ranked elections. SEEV systems have recently been developed for Condorcet voting [Ha22] and IRV elections [HBH24].

Our taxonomy and subsequent analysis highlight that it is increasingly challenging to design ranked e-voting systems as more information must be hidden from the public; this includes the plaintext votes as well as the components of the tally. It is desirable for any ranked election to reveal as little as possible through published information, however the advantages which are present for the solutions in one category may not be present for the solutions in

other categories that minimise information leakage and prevent the Italian attack. The most versatile systems for ranked e-voting currently belong to BAL-PUB; reliance on standard mix-based tallying means that many different ranked methods can be supported by a single design. Solutions in BAL-HID are much more limited due to their reliance on homomorphic tallying, and those in TAL-HID must design specialised counting procedures, for example through many secure MPC components [CGY22; DTY21; He21; TD24; WL04], to realise the same versatility offered by an e-voting solution in BAL-PUB. The major downside to the solutions in BAL-PUB is the threat of the Italian attack. Solutions belonging to BAL-HID mitigate against the Italian attack, whilst those belonging to TAL-HID not only mitigate against the Italian attack but also mitigate against any issues that may occur from revealing the tally. This includes embarrassment for choices that receive only a few votes and the weakening of mandates for choices that won the election with only a small margin of victory [Hu22].

Over all categories, we observe that it is challenging to tailor an e-voting solution for two particular cases: Condorcet-consistent C2 functions and STV. This is due to the complex tallying procedures present for each case. The work of Benaloh et al. [Be09] on the Shuffle Sum protocol is a promising solution with a novel method for representing and tallying STV ballots that avoids scalability issues present in related work [He07; Sm05]. However, Shuffle Sum is not E2E verifiable as it "is only concerned with tallying published, encrypted ballots" [Be09]. Harrison et al. [Ha22] present an e-voting solution for Condorcet voting that accounts for C2 functions whilst maintaining E2E verifiability and without reliance on trustworthy TAs, although their solution is limited to BAL-HID. The only work that provides a comprehensive overview of Condorcet-consistent C2 functions in TAL-HID is that of Hertel et al. [He21]. This work also maintains E2E verifiability, however their design for Schulze's method [Sc18] is not complete as it does not include an associated tie-breaking procedure. For some C2 functions including Schulze's method it is necessary to use a plaintext vote from the election itself in order to break any remaining ties between winners [Sc18; Ti87]. This is however at odds with the principle of vote privacy, as revealing this vote to break a tie would trivially violate the privacy of the individual who cast it. A key open problem in e-voting is the full implementation of these tie-breaking procedures for Condorcet-consistent C2 functions that maintains the privacy of the individuals who cast the tie-breaking votes.

Additionally, there is a lack of e-voting protocols belonging to TAL-HID that provide everlasting privacy or avoid the need for any trustworthy TAs. The e-voting protocols which aim to provide everlasting privacy for ranked methods are limited to BAL-PUB [CGG19; Co25; CPP13; LH15], whilst those which aim to provide self-enforcing e-voting are limited to BAL-HID [BAH19; Ha24; HBH24; Pa20a]. No solution aims to remove trustworthy TAs from STV elections. Minimising information leakage from the tallies, maintaining everlasting privacy, and removing the need for trustworthy TAs are all important aspects in the research of new e-voting systems that have been primarily considered independently of each other in the literature so far. A key open problem is the exploration of new e-voting

systems that maintain multiple or all of these properties for ranked elections alongside E2E verifiability.

We note that few solutions provide support for partial or equal rankings. Allowing a voter to omit choices in their ranking or to express equality between multiple choices enables them to be more sincere when casting their ballots. For particular elections, voters may find it easier to construct a ranking when they are not restricted to including every choice within it. Furthermore, the probability of a Condorcet winner existing within Condorcet elections has been found to be improved when equal rankings are permitted [FG80; Jo95; Va99]. Any e-voting system would benefit from the inclusion of partial rankings, as well as the inclusion of equal rankings for those that use Condorcet methods.

Finally, we see that few solutions allow votes to be cast through voting codes or fake credentials. This is largely due to the fact that many solutions do not specifically target remote voting and therefore assume a trusted voting device or assume that the voter is able to vote whilst keeping all credentials to themselves. No solution in TAL-HID, even those designed for remote voting [DTY21; He21; WB09], support either of these two primitives. A key open problem is whether such primitives can be adopted in order to reduce trust in the voting device and mitigate coercion whilst ensuring that the tallies remain hidden for ranked elections.

5 Conclusion

In this paper, we examined the design and security of ranked e-voting systems. We examined the literature for papers that aim to provide e-voting solutions for ranked voting and classified these according to the amount of information leakage maintained by each solution. Our taxonomy highlighted a current balance between the versatility of an e-voting solution and the strength of its privacy guarantees. Solutions which support a range of ranked methods often rely upon standard mix-based tallying and are susceptible to information leakage resulting from any published ballots or tallies. The solutions that adopt homomorphic tallying maintain stronger privacy guarantees, but are more tailored to specific ranked methods. Few works focus on adjacent concerns including support for partial or equal rankings, everlasting privacy, and the removal of trustworthy TAs.

Our taxonomy presents a number of open problems in e-voting for ranked elections. First, it is desirable to explore a full implementation of Condorcet-consistent C2 voting that wholly implements tie-breaking procedures whilst maintaining complete secrecy of the full tallies. Second, it is necessary to explore whether complete secrecy of the tallies is achievable alongside everlasting privacy and the removal of trustworthy TAs for ranked elections without compromises to E2E verifiability. Third, it is important to consider how assumptions necessary for remote elections, including trust in the voting device and use of credentials under coercion, can be weakened or removed for ranked elections whilst maintaining strong guarantees on privacy and verifiability.

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